

CS2422 Assembly Language & System Programming

September 9, 2003

Announcement

- Two sessions for this course:
 - CS2422-01 (張鈞法教授)
 - CS2422-02 (鍾葉青教授)

Textbooks

- Kip Irvine, “Assembly Language for Intel-Based Computers” 4th ed.
- Leland Beck, “System Software”
 - Using the first 4 chapters.
- Other good books:
 - Patterson & Hennessey, “Computer Organization and Design” (not Intel-based)
 - The Art of Assembly (freely available online)

Honor Code

- University code will be followed strictly to handle the cheating in assignments and exams.
- You are allowed (and in fact encouraged) to discuss the assignments, but the work must be your own.

Early Bonus & Late Penalty

- Early bonus: +2% for each day early, up to two days.
- Late penalty: -20% for each day (or partial day) late.

Why Learning Assembly?

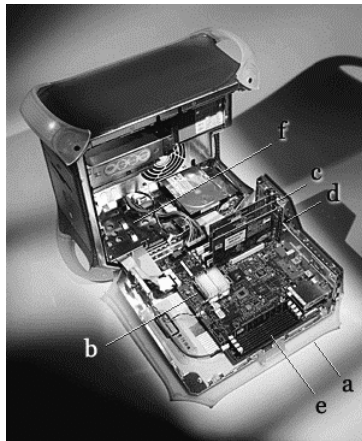
- A great way to learn how a computer really works:
 - To talk in the languages of the processors.
 - To see how a computer talks to the other devices.
- To build solid background for other courses:
 - Computer Architecture, Compilers, Operating Systems...etc.

What Exactly Is a PC?

A machine to...

- "...use email and surf the web" (Grandma)
- "...run MS-Office" (said Mom & Dad)
- "...play games and watch movies" (Kids)
- "...to write programs for Linux or Bill Gates' DOS/Windows" (CS Students)
- "...to show you the BIOS Screen" (EE Students)

Inside the case

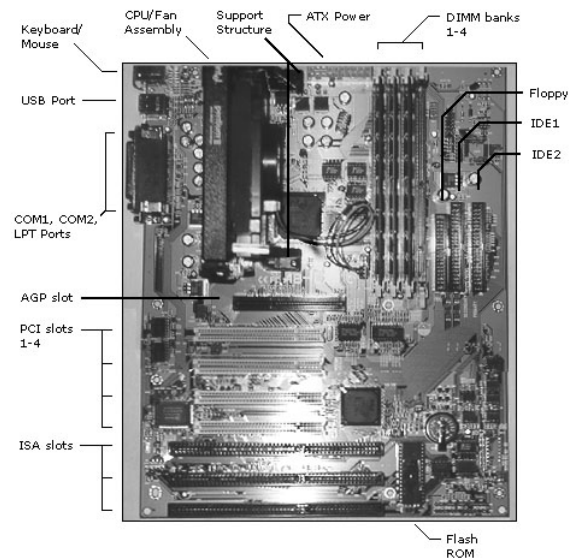


b. Processor

c. PCI slots (for I/O)

e. Memory slots

Motherboard



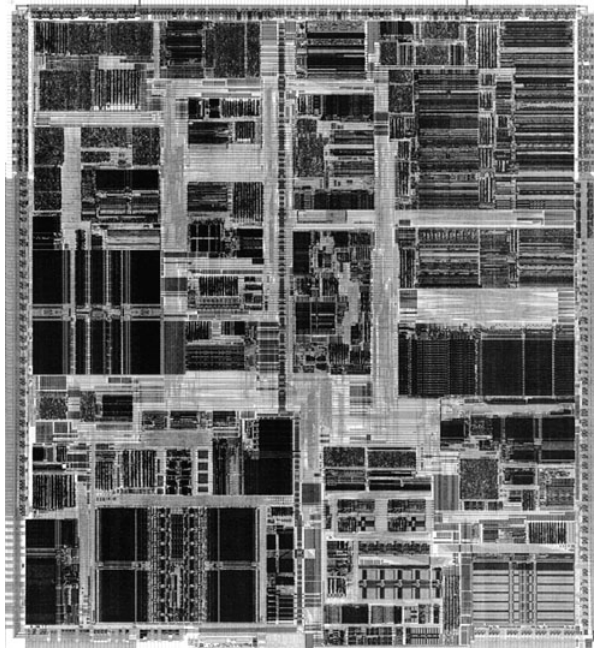
Memory



- RAM
- DRAM
- SRAM
- ROM
- Volatile / Non-Volatile
- Magnetic

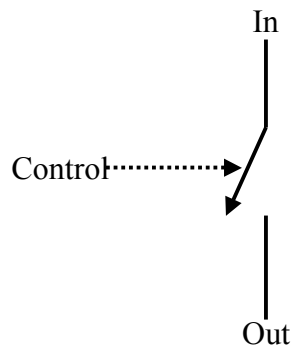
Processor

Pentium III Xeon



You only need switches and wires!

- Relays
- Vacuum tubes
- Transistors
- Integrated Circuits
- VLSI
- Nanotubes?
- Quantum Effect Devices?

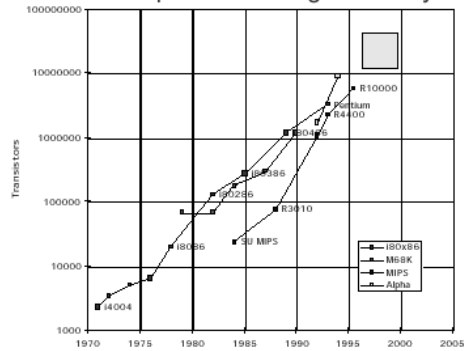


Improving Technology

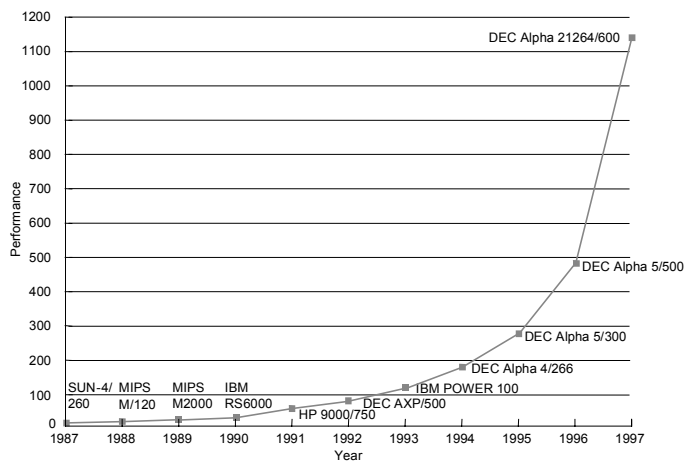
DRAM chip capacity

DRAM	
Year	Size
1980	64 Kb
1983	256 Kb
1986	1 Mb
1989	4 Mb
1992	16 Mb
1996	64 Mb
1999	256 Mb
2002	1 Gb

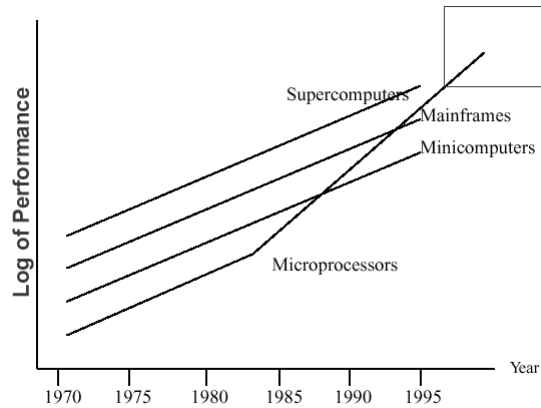
Microprocessor Logic Density



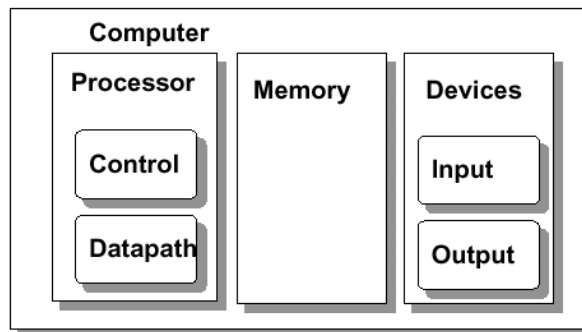
Performance Increase



Performance Trends



5 Classic Computer Components



Instruction and Operands

- Example:
MOV CX, 25
ADD AX, CX
- Data are in registers or memory.
- For now, we will focus on CPU, Registers, and Memory only.
- I/O will be discussed later.

Abstraction: C to ASM

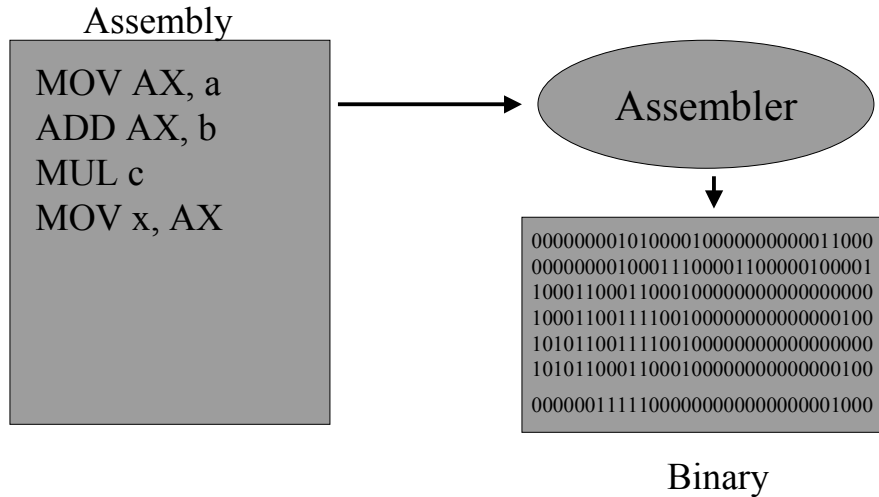
$x = (a+b) * b$

C compiler

Assembly

```
MOV AX, a
ADD AX, b
MUL c
MOV x, AX
```

Abstraction: ASM to Binary



Acknowledgement

- Many slides in this lecture are borrowed from Prof. Gary Bishop's COMP120 courses.